

```

import numpy as np
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 4
w,b = np.zeros(2),0

for _ in range(epochs):
    for x,y in zip(x_list,y_list):
        z = np.dot(x,w) + b
        ### condition area for y_pred
        y_pred = 1 if z >= 0 else 0
        E = y - y_pred
        print("error : ",E,"<- at epoch",_)
        w += E*x
        b += E
    print(w,b)

```

```

print("Prem·Massand,431")
print("Date:29/01/2026")

```

```

error :  -1 <- at epoch 0
[0. 0.] -1
error :  0 <- at epoch 0
[0. 0.] -1
error :  0 <- at epoch 0
[0. 0.] -1
error :  1 <- at epoch 0
[1. 1.] 0
error :  -1 <- at epoch 1
[1. 1.] -1
error :  -1 <- at epoch 1
[1. 0.] -2
error :  0 <- at epoch 1
[1. 0.] -2
error :  1 <- at epoch 1
[2. 1.] -1
error :  0 <- at epoch 2

```

```
import numpy as np
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 10
w,b = np.zeros(2),0

for _ in range(epochs):
    for x,y in zip(x_list,y_list):
        z = np.dot(x,w) + b
        ### condition area for y_pred
        y_pred = 1 if z >= 0 else 0
        E = y - y_pred
        print("error : ",E,"<- at epoch",_)
        w += E*x
        b += E
    print(w,b)

print("Prem Massand,431")
print("Date:29/01/2026")
```

```
... error :  0 <- at epoch 3
[2.  1.] -2
error :  0 <- at epoch 3
[2.  1.] -2
error : -1 <- at epoch 3
[1.  1.] -3
error :  1 <- at epoch 3
[2.  2.] -2
error :  0 <- at epoch 4
[2.  2.] -2
error : -1 <- at epoch 4
[2.  1.] -3
error :  0 <- at epoch 4
[2.  1.] -3
error :  0 <- at epoch 4
[2.  1.] -3
error :  0 <- at epoch 5
[2.  1.] -3
error :  0 <- at epoch 5
[2.  1.] -3
```

```

import numpy as np

x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])

epochs = 10

w = np.array([1, 1])
b = -1

for _ in range(epochs):
    for x, y in zip(x_list, y_list):
        z = np.dot(x, w) + b

        y_pred = 1 if z >= 0 else 0

        E = y - y_pred

        print("error :", E, "<- at epoch", _)

        w += E * x
        b += E

    print(w, b)

print("Prem Massand,431")
print("Date:29/01/2026")

```

```

... error : 0 <- at epoch 3
[2 2] -2
error : -1 <- at epoch 3
[2 1] -3
error : 0 <- at epoch 3
[2 1] -3
error : 0 <- at epoch 3
[2 1] -3
error : 0 <- at epoch 4
[2 1] -3
error : 0 <- at epoch 4
[2 1] -3
error : 0 <- at epoch 4
[2 1] -3
error : 0 <- at epoch 4
[2 1] -3

```

```

import numpy as np

x_list = np.array([[0,0],[0,1],[1,0],[1,1]])
y_list = np.array([0,0,0,1])
epochs = 10

w = np.array([1,1])
b = -1

for _ in range(epochs):
    total_error_in_epoch = 0
    for x, y in zip(x_list, y_list):
        z = np.dot(x, w) + b
        y_pred = 1 if z >= 0 else 0
        E = y - y_pred
        total_error_in_epoch += abs(E)
        print("error : ", E, "<- at epoch", _)
        w += E * x
        b += E
    print(w, b)

    if total_error_in_epoch == 0:
        print(f"\nConverged at epoch {_}. No errors found.")
        print("Final Weights:", w)
        print("Final Bias:", b)
        break

print("Prem Massand,431")
print("Date:29/01/2026")

```

```

... error : 0 <- at epoch 0
    [1 1] -1
    error : -1 <- at epoch 0
    [1 0] -2
    error : 0 <- at epoch 0
    [1 0] -2
    error : 1 <- at epoch 0
    [2 1] -1
    error : 0 <- at epoch 1
    [2 1] -1
    error : -1 <- at epoch 1
    [2 0] -2
    error : -1 <- at epoch 1
    [1 0] -3
    error : 1 <- at epoch 1
    [2 1] -2

```



```
import numpy as np

x_list = np.array([[0,0],[0,1],[1,0],[1,1]])

y_list = np.array([0,1,1,1])

epochs = 10

w = np.array([2,1])
b = -5

for _ in range(epochs):
    total_error_in_epoch = 0
    for x, y in zip(x_list, y_list):
        z = np.dot(x, w) + b
        y_pred = 1 if z >= 0 else 0
        E = y - y_pred
        total_error_in_epoch += abs(E)
        print("error : ", E, "<- at epoch", _)
        w += E * x
        b += E
    print(w, b)

print("Prem Massand,431")
print("Date:29/01/2026")
```

```
... error : 0 <- at epoch 3
[3 3] -2
error : 0 <- at epoch 3
[3 3] -2
error : 0 <- at epoch 3
[3 3] -2
error : 0 <- at epoch 3
[3 3] -2
error : 0 <- at epoch 4
[3 3] -2
error : 0 <- at epoch 4
[3 3] -2
error : 0 <- at epoch 4
[3 3] -2
```

b = -2

```
for _ in range(epochs):
    total_error_in_epoch = 0
    for x, y in zip(x_list, y_list):
        z = np.dot(x, w) + b
        y_pred = 1 if z >= 0 else 0
        E = y - y_pred
        total_error_in_epoch += abs(E)
        print("error : ", E, "<- at epoch", _)
        w += E * x
        b += E
        print(w, b)

    if total_error_in_epoch == 0:
        print(f"\nConverged at epoch {_}. No errors found.")
        print("Final Weights:", w)
        print("Final Bias:", b)
        break

print("Prem Massand,431")
print("Date:29/01/2026")
```

```
... error : 0 <- at epoch 0
[2 1] -2
error : 1 <- at epoch 0
[2 2] -1
error : 0 <- at epoch 0
[2 2] -1
error : 0 <- at epoch 0
[2 2] -1
error : 0 <- at epoch 1
[2 2] -1
error : 0 <- at epoch 1
[2 2] -1
error : 0 <- at epoch 1
[2 2] -1
error : 0 <- at epoch 1
[2 2] -1
```

```
Converged at epoch 1. No errors found.
Final Weights: [2 2]
Final Bias: -1
Prem Massand,431
Date:29/01/2026
```



```
import numpy as np
```

```
x_list = np.array([[0,0],[0,1],[1,0],[1,1]])  
y_list = np.array([0,1,1,0])
```

```
epochs = 10
```

```
w = np.array([3,4])  
b = -5
```

```
for _ in range(epochs):  
    total_error_in_epoch = 0  
    for x, y in zip(x_list, y_list):  
        z = np.dot(x, w) + b  
        y_pred = 1 if z >= 0 else 0  
        E = y - y_pred  
        total_error_in_epoch += abs(E)  
        print("error : ", E, "<- at epoch", _)  
        w += E * x  
        b += E  
        print(w, b)  
  
    if total_error_in_epoch == 0:  
        print(f"\nConverged at epoch {_}. No errors found.")  
        print("Final Weights:", w)  
        print("Final Bias:", b)  
        break  
  
print("Prem Massand,431")  
print("Date:29/01/2026")
```

```
*** error :  0 <- at epoch 0  
[3 4] -5  
error :  1 <- at epoch 0  
[3 5] -4  
error :  1 <- at epoch 0  
[4 5] -3  
error : -1 <- at epoch 0  
[3 4] -4  
error :  0 <- at epoch 1  
[3 4] -4  
error :  0 <- at epoch 1  
[3 4] -4  
error :  1 <- at epoch 1  
[4 4] -3
```